

The Last Writer Wins: Installing Epistemic Disposition in Multi-Agent LLM Pipelines

Sam Bobo

Draft v0.4 — July 2026 | Target: arXiv cs.CL

Abstract

Multi-agent LLM pipelines route work through specialist models and aggregate their outputs through a synthesizer. We study *disposition*—what a model chooses to emit independent of what it knows, operationalized as five epistemic behaviors (training-cutoff disclosure, modeled-assumption flagging, precise vocabulary distinctions, jurisdictional distinguishing, and hedging)—and ask where disposition in a pipeline comes from and whether it can be installed. Across ~ 630 audited runs of a four-model “Council of Experts” on consumer hardware, we find: (1) the pipeline architecture itself lifts disposition density $3\text{--}9\times$ over single-shot use, model-agnostically, and *amplifies* under simultaneous behavioral demand where single-shot *dilutes*; (2) upstream installation fails robustly: prompting and supervised fine-tuning on behavior-rich exemplars produce large seat-level lifts ($\approx 2\times$, bootstrap CIs excluding baseline) that are indiscriminate—both hedge on a trigger-free control case—and that vanish from the pipeline’s final output, while reference-free preference optimization installs no seat-level magnitude across three model lineages (Mistral, Llama-3, Qwen-3), invariant to a $3.2\times$ increase in preference data. Preference optimization’s one real effect is subtractive and conditional: on the weakly-aligned legal seat it suppresses unwarranted hedging below baseline—a finding that survives triangulation across three instruments (pattern-based, NLI entailment, and unanimous dual LLM judges) and generalizes across objectives (ORPO and CPO), though only ORPO suppresses *selectively*; the effect has no purchase on an already-aligned seat and did not replicate on a third lineage. Triangulation also caught one finding produced by the primary metric alone (an apparent hedging *increase* on the finance seat) and retired it as instrument artifact; (3) the mechanism is not sentence-level content entanglement (refuted, $r = -0.10$) but a **synthesizer register**: each aggregator model writes at its own characteristic disposition density, nearly independent of seat input—over-dense input can *invert* output density—while the synthesis prompt’s preservation instructions act as a *graded* gain control: output density rises monotonically with the number of preservation clauses on all three synthesizers tested ($2\text{--}3\times$ full-scale). The register also survives direct attack — ORPO-training the synthesizer itself on synthesis-level pairs moves its band no more than seat training moved the seats, and heating one, two, or all three seats adds nothing at the pipeline’s mouth (non-additive, $\rho = -0.80$). Weight-level training fails at every locus; **the accessible control surface is the last writer’s instructions**. Practically: a pipeline’s epistemic posture is governed by its last writer and that writer’s instructions; upstream installation is mostly futile. The evidence base: 628 pre-registered, audited runs across three model lineages, six installation arms, and both pipeline loci (seats and the synthesizer itself), scored under three independent instruments with unanimous judge agreement on the surviving positive claim—all \$0-budget and fully reproducible on a single 32 GB consumer machine.

1 Introduction

Production LLM systems increasingly take the form of pipelines: multiple models, often specialized, whose outputs are aggregated by a final model into the answer a user sees. Evaluation practice, however, remains

seat disposition varies with arm
(prompting / SFT / ORPO / CPO)

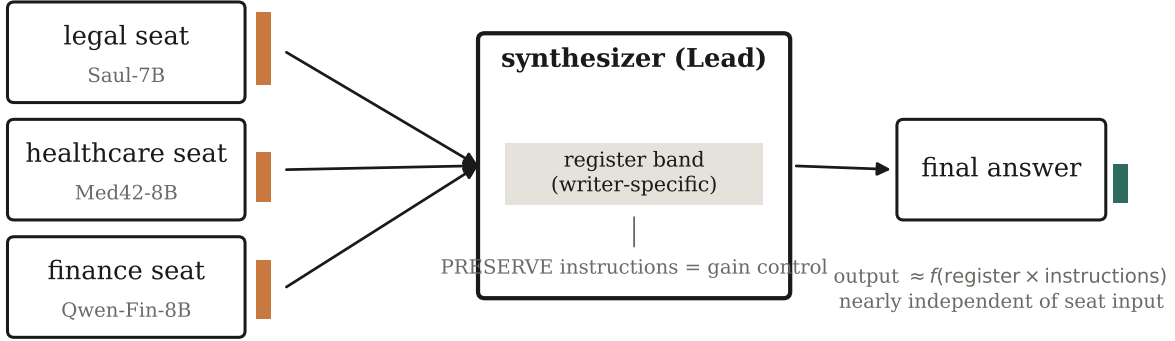


Figure 1: The last writer wins. Specialist seats emit disposition at densities that vary with the installation arm (orange bars), but the synthesizer normalizes everything to its own characteristic *register band*; the synthesis prompt’s PRESERVE instructions act as a gain control on that band. Final-output disposition (green) is nearly independent of what the seats emit (§6, Fig. 3).

largely single-model: we measure whether a *model* hedges appropriately, discloses its limits, or labels assumptions. This paper asks what happens to those behaviors *inside a pipeline*—who originates them, whether they can be installed, and whether they survive aggregation.

We distinguish two evaluation axes that are conflated in practice. **Content** is whether the right substantive material appears (measured here by per-case expert rubrics). **Disposition** is what a model chooses to emit about its own claims. Our experiments show these axes are orthogonal—the model family that wins content loses disposition and vice versa—and that pipeline disposition is governed almost entirely by the final aggregation stage.

Contributions. (i) A two-axis evaluation of a heterogeneous local specialist pipeline, with a composite disposition metric (CDS) and an architectural lift ratio (ALR) (§4). (ii) A controlled comparison of installation mechanisms—prompting, exemplar SFT, and reference-free preference optimization (ORPO and CPO)—under pre-registered predictions, with a trigger-free control case that separates *responsive* from *habitual* behavior, replicated across three seats of distinct model lineage (§5). (iii) Identification, ablation, and stress-testing of the **synthesizer register** as the mechanism governing which installed behaviors reach the pipeline’s output: it survives direct ORPO training of the synthesizer, seat input is non-additive, and the synthesis instructions form a measured, monotone gain control (§6). (iv) A three-instrument measurement-validation protocol (pattern matching, calibrated NLI entailment, blinded pairwise LLM judges) that both confirms the surviving claims and catches a false finding produced by the primary metric (App. E). (v) A fully local, \$0, consumer-hardware protocol with append-only audit logs for all 628 runs.

2 Related work

Multi-agent orchestration. Debate and role-played collaboration improve factuality and reasoning [1, 2, 3, 4], and recent work compares orchestration against the strongest single model [5] or trains orchestrators end-to-end [6]. This literature measures content outcomes. The closest work on the *aggregation* step, Parallel-Synthesis [7], trains a synthesizer adapter and explicitly distills reasoning behavior from text-concatenation synthesis—an implicit acknowledgment that naive aggregation loses behavior, which our reg-

ister mechanism explains and quantifies for epistemic behavior specifically. We measure what orchestration does to epistemic behavior and find an amplification effect independent of the model filling the seats.

Specialists vs. generalists. Evidence is mixed on whether small domain fine-tunes beat larger generalists in-domain [8, 9]. Our content results replicate the negative side (a 20B open MoE outperforms a four-specialist council 42% $\rightarrow$ 25–31% on rubric coverage) and motivate the pivot: the surviving specialist value is dispositional, not informational.

Uncertainty expression and abstention. Verbalized uncertainty [10], linguistic calibration [11], knowing-what-you-know [12], abstention training [13] and recent confidence-aware abstention work [14, 15] train and evaluate single models end-to-end. A parallel line asks whether epistemic markers *faithfully* track a model’s internal confidence—and finds systematic deficiencies [16, 17, 18]. Our contribution is orthogonal to faithfulness: we take marker *emission* as the observable and ask where it originates and survives in a pipeline, not whether it is calibrated to hidden confidence. A seat’s trained uncertainty behavior must survive an aggregator, and mostly does not. Two adjacent 2026 lines make the pipeline dimension timely: “confidence laundering” [24] names, as a position, the loss of decision uncertainty at agent interfaces and argues for a latent uncertainty carrier; and a quantitative line propagates *numeric* confidence estimates through multi-step agent decisions [25]. We complement both empirically and from the opposite direction: we measure *emitted linguistic disposition*, show its loss is not an interface accident but a property of the last writer (the register), and show that upstream training cannot push behavior through it.

Preference optimization for behavior. DPO [19] and ORPO [20] are standard for style/behavior shaping; DPO’s length bias is documented [21], which our pair-construction controls address. Preference-data scaling is itself contested: simply adding more samples per prompt often fails to improve and can degrade downstream behavior [22, 23]—consistent with our dose-invariance null, which we further attribute to a downstream aggregation ceiling rather than to the optimizer. Comparisons of system-prompt steering vs. preference training exist for single models; we compare them *through* a pipeline.

3 Apparatus

Council. Four local models via Ollama on a 32 GB MacBook Air M5: Phi-4-14B Lead (planner + synthesizer), and three specialist seats—Llama3-Med42-8B (healthcare), Saul-7B-Instruct-v1 (legal), Qwen-Open-Finance-R-8B (finance), all Q4/Q8 GGUF. The Lead performs step-back decomposition [29] and dispatches self-contained sub-questions; seats never see the original query (this architectural choice, not prompt strengthening, eliminated cross-domain lane bleed at sub-10B scale). The Lead then synthesizes under a structurally enforced Tensions-then-Synthesis prompt containing four PRESERVE instructions.

Cases. Seven three-domain scenarios: five failure-mode-targeted (synthesis stress, jurisdictional vocabulary, quantitative discipline, recency honesty, adversarial cross-domain tension) plus a matched disposition pair—case 6 (*trigger-heavy*: demands all five behaviors simultaneously) and case 7 (*trigger-light*: an organizational-strategy question warranting none). Case 7 is the **responsiveness gate**: any arm emitting disposition there is behaving habitually, not responsively.

Scale. 462 imported, audited runs (append-only JSON, one file per run, each carrying per-phase inputs/outputs/backends); all inference via Ollama on one fanless M5 MacBook Air, 32 GB unified memory, 26.8 GB Metal working set. Runs are nondeterministic across repeats even at temperature 0 (runtime batching); we treat repeats as seeds and report bootstrap intervals.

Metrics. Behavior density = pattern-matched occurrences of the five behavior families per 1,000 characters. $CDS = \text{density} \times \sqrt{\text{distinct behaviors}/5}$. $ALR = \text{council density}/\text{matched single-shot density}$. Seat-level density is computed on the specialist’s own turn from the audit log; final density on the synthesized output.

Table 1: Installation arms on the legal seat (cell-2 variance pass; $n = 5$ seeds $\times$ 7 cases per arm, bootstrap 95% CIs over run-level observations). Bold marks CIs clear of baseline (seat) and the best value per column.

Arm	Seat density	Final-output CDS	Trigger-light gate
A' baseline	0.89 [0.69, 1.11]	0.86 [0.64, 1.08]	0.96
Prompting	1.85 [1.42, 2.32]	0.59 [0.40, 0.81]	3.03 ✗
SFT-on-chosen	1.77 [1.46, 2.09]	0.58 [0.43, 0.73]	1.21 ✗
ORPO	0.87 [0.60, 1.17]	0.66 [0.49, 0.85]	0.15 ✓

(Definitions and regexes are versioned with the code; see §8.)

Installation arms (legal seat, single-variable design, seed 42): A' = conversion-control baseline (identical Saul weights through our fp16→GGUF→Q4 pipeline); B = a frozen behavior-spec addendum on the seat's system prompt (five behaviors, one example each, an explicit "do not force it" clause); SFT = LoRA on 91 behavior-rich *chosen* responses; ORPO = LoRA reference-free preference optimization on the same 91 chosen responses paired with behavior-stripped rejections. Pairs are content-controlled (chosen/rejected are rewrites of the same base answer), length-ratio filtered (0.8–1.4) against length bias, overlap-filtered (Jaccard ≥ 0.35 on capitalized tokens), and leakage-screened against all seven cases. Replication arms repeat the recipe with one variable changed: ORPO on dose-matched 91-pair sets for the healthcare seat (Med42, preference-aligned) and finance seat (Qwen-Open-Finance, SFT-only), a CPO arm on the identical legal pairs (only the objective differs), and an ORPO arm on the *synthesizer itself* (Qwen2.5-7B Lead, 88 synthesis-level pairs built from recorded deliberations; sequence length 4,096 — an amendment inherent to the locus, since syntheses integrate all seat outputs and no synthesis example fits the seat arms' 1,792). All protocol amendments were documented before training; predictions were pre-registered before each cell ran.

4 Result 1: architecture shapes disposition

Across seven cases, council configurations emit 3–9 $\times$ the behavior density of matched single-shot baselines. Aggregate CDS: specialists-v2 0.928, Opus-as-council 0.669, specialists-v1 0.584, gpt-oss-as-council 0.460, Opus single-shot 0.159, gpt-oss single-shot 0.099. ALR: Opus pair 2.99 $\times$, gpt-oss pair 3.91 $\times$, local v1 5.14 $\times$, upgraded v2 8.77 $\times$ —the synthesis prompt's preservation scaffold does real work even with no specialist alignment in the seats. Under the trigger-heavy case the asymmetry sharpens: council modes rise to 1.66–1.82 $\times$ their per-case baseline while single-shot falls to 0.68 $\times$ —orchestration *amplifies* disposition under simultaneous demand where a single context *dilutes* it. On the trigger-light case every mode, specialist or frontier, emits zero: measured disposition is responsive, not habitual, across the board.

5 Result 2: installing disposition — magnitude vs. responsiveness

Figure 2 and Table 1 give the installation triangle at $n = 5$ seeds per cell (140 runs; the CPO arm, below, is included in the figure for comparison):

Prompting and SFT install real seat-level magnitude (CIs exclude baseline)—and both fail the gate, hedging on the trigger-free case (prompting despite its explicit "do not force it" clause). Both also lose their gains at the pipeline output: final CDS lands at or below the untrained baseline. ORPO installs no significant seat-level magnitude; its effect is qualitative—it is the only arm that *suppresses* unwarranted hedging below

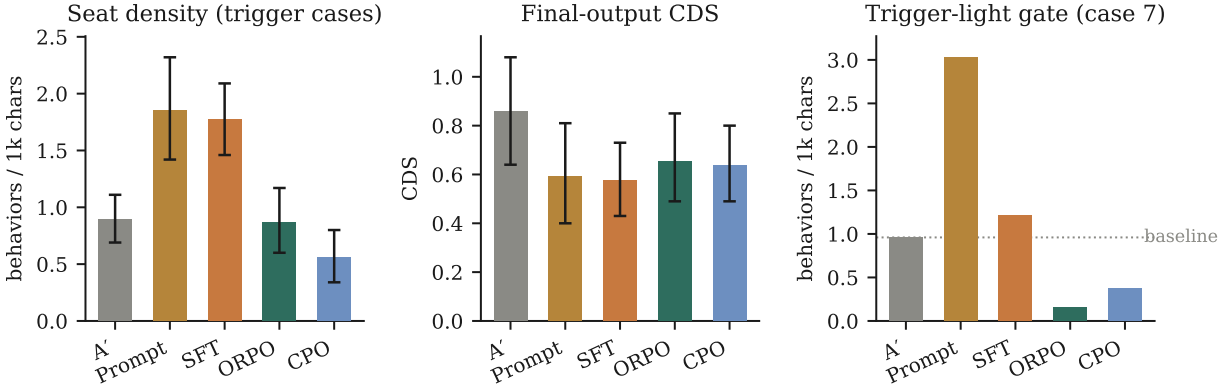


Figure 2: The installation triangle. Prompting and SFT install real seat-level magnitude (left) that synthesis strips (center) and that fires on the trigger-free control (right); preference optimization installs nothing and suppresses the gate. Bars: means over $n = 30$ run-level observations (6 trigger cases $\times$ 5 seeds); whiskers: bootstrap 95% CIs (10^4 resamples); gate panel: $n = 4\text{--}5$ per arm, dotted line = untrained baseline.

baseline while leaving trigger-case behavior intact and imposing no content tax (rubric coverage identical to A').

The null is dose-invariant. Retraining ORPO at $3.2\times$ dose (292 content-controlled pairs, epoch-matched to the 91-pair run) moved seat-level magnitude by zero: seat density 0.84 [0.58, 1.11] vs. the 91-pair 0.85 and baseline 0.84, even though the larger set raised training-time preference accuracy from 0.48 to 0.94. More preference data was internalized but not emitted—strong evidence that ORPO’s seat effect is not gradual installation of magnitude but a bounded suppression of unwarranted behavior. (The suppression itself weakened slightly with dose: the trigger-light gate rose from 0.15 to 0.49, still below baseline; more diverse pairs modestly broadened where the model deploys hedging.)

The null generalizes across lineages; the suppression does not. Pre-registered replications trained ORPO identically on two further seats. On the healthcare seat (Med42, already preference-aligned): no magnitude (1.40 [1.15, 1.68] vs. A' 1.30 [1.07, 1.54]) and no responsiveness effect to detect—the untrained baseline already emits 0.00 on the trigger-light case (dispatched in all runs); a pre-aligned model leaves no miscalibration slack to suppress. On the finance seat (Qwen-Open-Finance, SFT-only, Qwen-3): no magnitude (1.32 vs. 1.37, CIs overlapping) and *no suppression* (gate 0.81 $\rightarrow$ 1.21 under the pattern metric)—the replication prediction was falsified. The suppression is therefore conditional on base-model miscalibration and not guaranteed even then.

The suppression is method-general; the selectivity is ORPO’s. A CPO arm [26] on the identical legal pairs (only the objective changed) reproduces the gate suppression under both instruments (0.37 pattern / 0.45 NLI-cutoff vs. A'’s 0.96 / 1.06)—but CPO also dampens *warranted* trigger-case hedging (0.89 $\rightarrow$ 0.56), where ORPO leaves it intact (0.87). Gate-to-trigger selectivity: A' 1.08, CPO 0.66, ORPO 0.17. Suppression is a property of reference-free preference optimization on these pairs; suppressing *only where hedging is unwarranted* is, in our data, ORPO-specific.

The null extends to the last writer. If the register is merely where disposition happens to live, training the synthesizer itself should move it. It does not: ORPO on the Lead (same recipe, 88 dose-matched synthesis-level pairs, seats held at their untrained production versions) leaves the pipeline-mouth band unmoved — final density 0.89 [0.67, 1.11] vs. the conversion control’s 1.03 [0.79, 1.28], against a pre-registered threshold of $|\Delta| \geq 0.25$ with disjoint CIs. The control band also reproduces this writer’s \$6 register (1.03 vs. 1.24),

Table 2: Final-output density by synthesizer $\times$ prompt $\times$ seat-input arm (6 cases each; seat input densities in Fig. 3).

Synthesizer	PRESERVE, base seat	PRESERVE, 2–3 $\times$ hot seat	no-PRESERVE, base	no-PRESERVE, hot
Phi-4-14B	1.17	0.61	0.52	0.31
gpt-oss-20B	0.63	0.86	0.20	0.13
Qwen2.5-7B	1.24	1.06	0.24	0.62

stable across cells. The trigger-light gate was already near-floor untrained (0.15), so no suppression is claimed — the same no-slack situation as the pre-aligned seat. Weight-level training now fails at *every* locus tested.

Instrument validation. Because these verdicts rest on measurement, we validated the instrument stack (§E): a calibrated NLI entailment scorer (chosen-vs-rejected AUC 0.929 on the construction-labeled pair corpus) and blinded, order-swapped, evidence-grounded pairwise LLM judges [28]. The magnitude nulls and healthcare zeros are instrument-robust (pattern + NLI); the legal suppression is three-instrument robust (pattern, NLI-cutoff 1.06 $\rightarrow$ 0.35, and 14/14 decided judge pairs unanimous under both judges, sign-test $p = 0.016$ each, no length bias); and the apparent finance *reversal* (0.81 $\rightarrow$ 1.21) appeared under the pattern metric only—NLI shows no increase on any family—and is retired as instrument artifact. The finance result stands as a non-replication (no suppression), not a worsening.

Two further details. First, prompting was tested at two loci: on the generalist single-shot (a $6\times$ final-output lift, 0.099 $\rightarrow$ 0.589, that failed the gate at 0.985—its answer to the hybrid-work communication question opened a section “Current State Snapshot (as of my training data)” and closed with “actual results may vary”: disclosure theater where nothing warranted disclosure) and on the seat (the table above). Second, content is unaffected by successful installation: rubric coverage is 12/36 for both A’ and ORPO (9/36 for the original production seat)—no content tax. Gate cells use $n = 4\text{--}5$ (the planner routes no specialists on the trigger-light case in some runs; routing variability is itself Ollama-runtime nondeterminism at temperature 0). Summary: *prompting and exemplar training change what a model says, loudly and indiscriminately; preference training installs nothing—at best it removes miscalibrated hedging where the base model has it, ORPO selectively and CPO bluntly, and only the last writer decides what survives.*

6 Result 3: the synthesizer register

Why do installed behaviors vanish? A pre-registered entanglement hypothesis—content-woven markers survive aggregation, detached boilerplate is stripped—was **refuted**: entangled share is flat across arms (baseline 56%, prompting 61%, ORPO 61%, SFT 67%—SFT’s markers are the *most* content-woven yet retain worst), retention splits by arm not position (baseline 1.08, ORPO 0.96 vs. prompting 0.49, SFT 0.49), and entanglement does not predict retention ($r = -0.10$ over 102 runs). Notably the refuted hypothesis and the overturned PRESERVE prediction (below) were both pre-registered—we report them as part of the record.

A 72-run ablation (2 input arms $\times$ 3 synthesizer models $\times$ PRESERVE instructions on/off $\times$ 6 cases) located the mechanism (Fig. 3, Table 2):

Each synthesizer writes at its own characteristic density band regardless of input (registers are writer-specific); over-dense input does not transmit and can *invert* output density (Phi-4: 1.17 $\rightarrow$ 0.61—over-

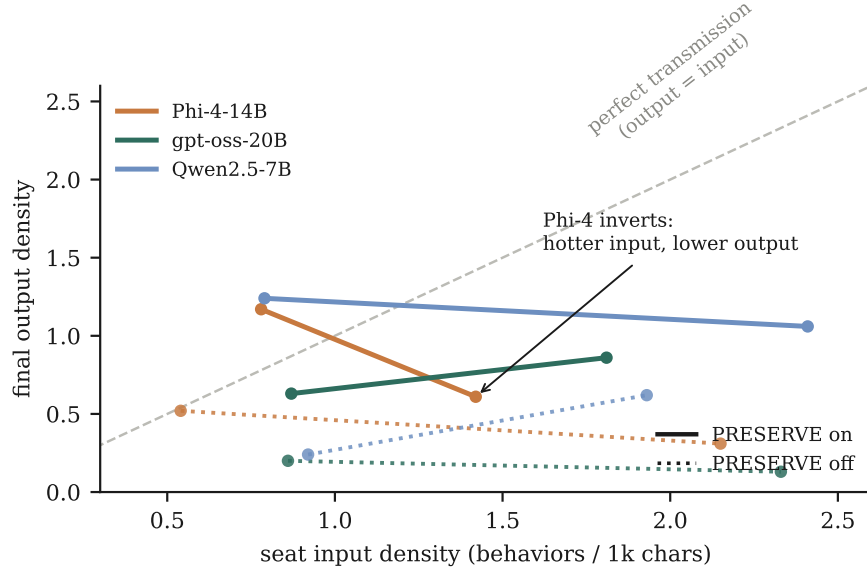


Figure 3: The synthesizer register (72-run ablation). Every configuration sits far below the perfect-transmission diagonal; slopes are nearly flat (input density does not transmit) and Phi-4’s is negative (overdense input *inverts* output). Solid vs. dotted separates PRESERVE instructions on/off—a 2–5 $\times$ gain on every Lead. Points: mean final-output density over 6 cases per configuration.

correction, saturated hedging read as stylistic noise); and the PRESERVE instructions are a **gain control**, their removal collapsing output 2–5 $\times$ on every synthesizer (overturning our registered prediction that instructions would not matter). Mechanism: *final disposition* $\approx f(\text{register} \times \text{instructions})$, *nearly independent of seat input*. ORPO “survives” by staying inside the register; prompting and SFT “strip” by exceeding it. The dose-invariance of §5 is the register’s corollary: if the aggregator’s band, not the seat’s input, sets output disposition, then no amount of additional preference data at the seat can raise it—which is exactly what 3.2 $\times$ dose showed—and why the seat replications found no pipeline-level effect to transmit.

Two further ablations sharpen the mechanism (Fig. 4). First, the binary PRESERVE on/off contrast generalizes to a *graded, monotone* response: retaining $k \in \{0, 1, 2, 3\}$ preservation clauses raises final-output density monotonically on every Lead—instructions are a measured gain control, not a switch. Second, disposition is *non-additive* in seat input: applying the behavior-spec override to one, two, or all three seats leaves the pipeline mouth flat ($\rho = -0.80$, spread 0.38, overlapping CIs)—closing the remaining alternative that seat disposition partially accumulates and is diluted. Combined with the Lead-training null (§5), the mechanism statement finalizes as: *final disposition* $\approx f(\text{register} \times \text{instructions})$, *robust to weight-level training at either locus; the accessible control surface is the last writer’s instructions*.

7 Background: why disposition, not content

The council was built to test whether small domain fine-tunes beat generalists on content. They do not: a single 20B open-weights MoE (gpt-oss-20B) outperforms the full council on rubric coverage (42% vs. 25%; 31% after best-available specialist upgrades), and the council exhibited classic small-model failures (confabulated trial citations, repealed regulation treated as live). What specialists retained was disposition—the highest CDS of any configuration—which motivated the installation question this paper answers. We

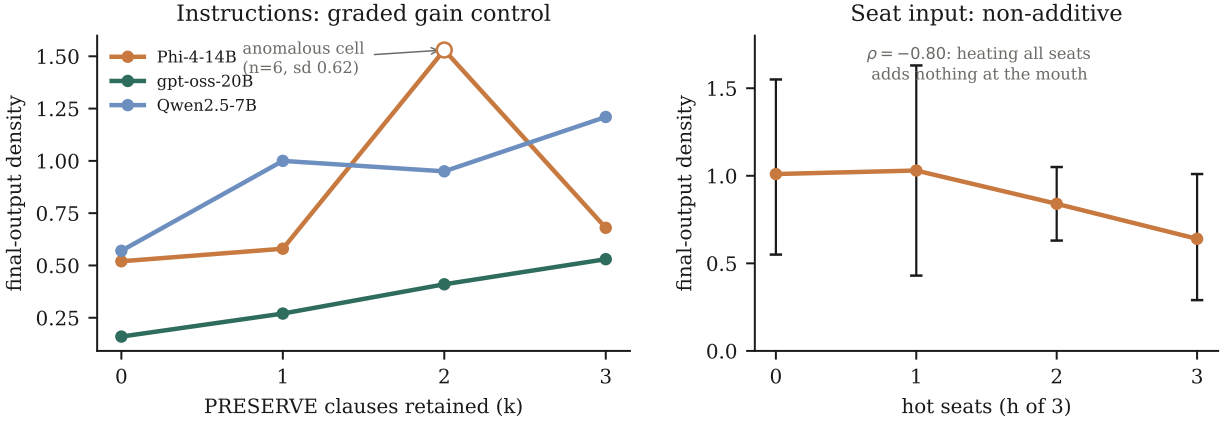


Figure 4: The register’s two stress tests (96 runs). *Left*: final-output density rises monotonically with the number of PRESERVE clauses retained in the synthesis prompt, on all three Leads (Spearman $\rho = +1.00/+0.80/+0.80$; spans $3.3\times$ and $2.1\times$ for gpt-oss and Qwen). Phi-4’s $k=2$ cell (open marker) is anomalous—no other Lead bumps there and its within-cell dispersion equals its between-level differences at $n=6$ —and is reported as probable noise. *Right*: heating h of 3 seats with the behavior-spec override leaves pipeline-mouth density flat-to-declining ($\rho = -0.80$; CIs overlap): seat disposition is not partially additive and then diluted, it does not transmit. Points are means over 6 cases; whiskers bootstrap 95% CIs.

report this as motivating context, not a contribution.

8 Limitations

Pattern-based scoring has paraphrase blind spots and rewards concision; we mitigate with per-claim-normalized NLI scoring and pairwise LLM judges (§E), but the validation rests on inter-instrument convergence rather than a human-rated anchor, two NLI hypothesis families calibrated degenerately (documented), and the judges’ evidence-grounding gate rejected most pairs on dense numeric finance text. ORPO substitutes for DPO due to a memory defect in the only local DPO trainer available (documented); installation claims are scoped to reference-free preference optimization (ORPO, CPO). Suppression evidence concentrates on one weakly-aligned seat; trigger-light gate cells use $n = 4-5$. The synthesizer-training null rests on one Lead (Qwen2.5-7B; a 14B Lead exceeded the training budget) at 88 pairs vs. the seat arms’ 91; the gain-curve and additivity cells use $n = 6$ per point without seed repeats, and Phi-4’s anomalous $k=2$ cell is unresolved (reported as probable noise; a seeded replication would settle it). A preference-data range of 91–292 pairs (dose-invariant, but far larger scale untested), $n = 5$ seeds, seven self-authored cases, and three local synthesizers $\leq 20B$ —frontier aggregators untested. All experiments ran on one consumer machine at \$0 API cost; we regard the reproducibility this buys as partial compensation for the scale it forgoes.

9 Conclusion

In multi-agent pipelines, epistemic disposition is not additive across stages: it is set by the last writer’s register and that writer’s instructions. Upstream installation by prompting or exemplar tuning produces loud but indiscriminate behavior that the aggregator strips; preference optimization installs nothing at all—across three model lineages and a $3.2\times$ data dose—because the aggregator’s register, not the seat, sets the

ceiling. Its one dependable effect is subtractive: removing miscalibrated hedging where the base model has it, selectively under ORPO, bluntly under CPO, and on none of our seats reliably enough to treat as a design tool. The register itself withstands direct training of the synthesizer, and seat input is non-additive at the mouth—but the synthesis instructions form a monotone, $2\text{--}3\times$ gain control. In a multi-agent pipeline, the accessible control surface for epistemic disposition is not anyone’s weights; it is the last writer’s instructions. Builders who want calibrated pipelines should tune the synthesizer’s instructions and choose the final model deliberately—and evaluate disposition at the pipeline’s mouth, not the seat.

Reproducibility. All code, prompts, 628 imported audited runs, pre-registrations, protocol amendments, verdicts (including refuted hypotheses), the pair datasets for all three seats, LoRA adapters, the NLI and judge instrument harnesses, and regeneration scripts: <https://github.com/srbobo/council-of-experts-research>.

A Glossary

Seat

one specialist model in the council, answering only its dispatched sub-question (e.g., Saul-7B = the legal seat).

Synthesizer / Lead

the model that receives all seat outputs plus the original question and compresses them into the final answer—the pipeline’s last writer.

Disposition

what a model chooses to emit independent of what it knows, operationalized as five behavior families: (1) training-cutoff disclosure; (2) modeled-assumption flagging; (3) precise vocabulary distinctions; (4) jurisdictional distinguishing; (5) **hedging**—stated conditionality of a claim (“this may vary if...”), *not* refusal or vagueness.

Synthesis stripping

behavior density present at a seat’s output but absent from the final output after synthesis.

Synthesizer register

a Lead model’s characteristic output disposition density—writer-specific and largely independent of seat input (§6).

Responsive vs. habitual

responsive behaviors appear only when domain triggers warrant them (trigger-light gate, case 7); habitual behaviors fire regardless.

Alignment

post-pretraining procedures (SFT, RLHF, DPO/ORPO/CPO) shaping disposition, as distinct from the pretraining corpus, which shapes knowledge.

B Model selection

Every selection cleared seven gates: (1) GGUF availability on a mirror Ollama can pull; (2) research-permissive license; (3) ChatML-compatible or locally fixable chat template; (4) recognized-lab provenance;

(5) distinct lineage per seat (four labs, three model families—a family-level training artifact cannot masquerade as a council effect); (6) active maintenance; (7) evidence of third-party use.

Memory budget. Sequential mode holds the Lead warm across all five phases, so peak $\approx M_{\text{lead}} + \max_{\text{seat}} M_{\text{seat}} + \text{KV} + \text{OS reserve} \leq 32 \text{ GB}$; Metal’s recommended working set on the test machine is 26.8 GB. This bounds the Lead at $\leq 14\text{B}$ (Q4) and seats at $\leq 8\text{B}$.

Lead: Phi-4-14B (MIT)—best reasoning-per-parameter at 14B for structured planner output; Qwen2.5-14B rejected on memory headroom, Llama-3.1-8B on synthesis capacity, 22–27B dense models on the Metal cap. The Lead is deliberately *not* domain-tuned. **Healthcare: Llama3-Med42-8B**—the only 8B clinical fine-tune with a multi-stage *preference-alignment* stage. **Legal: Saul-7B-Instruct-v1**—the only sub-13B continued pretrain (30B+ tokens) on multi-jurisdiction common-law text. **Finance: Qwen-Open-Finance-R-8B**—newest backbone of any seat (Qwen3), $> 50\%$ finance corpus. **MoE comparator: gpt-oss-20B**— $\sim 3.6\text{B}$ active parameters, reasoning-tuned, Apache-2.0; fits beside the Lead ($14 + 9 = 23 \text{ GB} < 26.8 \text{ GB}$), unlike Qwen3-30B-A3B (27 GB) or Mixtral-8x7B (35 GB).

C Metrics and estimation

Let B be the five behavior families and P_b the regex pattern set for family b . For text t :

$$\begin{aligned} \text{Density} \quad d(t) &= \frac{1000}{|t|} \sum_{b \in B} \sum_{p \in P_b} |\text{matches}(p, t)|, \\ \text{Breadth} \quad k(t) &= \frac{1}{|B|} \left| \{ b \in B : \exists p \in P_b, \text{matches}(p, t) \neq \emptyset \} \right|, \\ \text{CDS} \quad \text{CDS}(t) &= d(t) \cdot \sqrt{k(t)}, \\ \text{ALR} \quad \text{ALR}_m &= \bar{d}_{\text{council}}(m) / \bar{d}_{\text{single}}(\hat{m}), \\ \text{Retention} \quad R &= \text{clip}(d(\text{final})/d(\text{seat}), 0, 2). \end{aligned}$$

The square root in CDS softly penalizes narrow emission without letting one absent family zero the score (a geometric mean was rejected for exactly that failure); $\alpha = \frac{1}{2}$ is a design choice, sensitivity unreported. Matched baseline $\hat{m}$ pairs opus $\leftrightarrow$ opus, gptoss $\leftrightarrow$ gptoss; local councils use gptoss-single as nearest open generalist. *Worked example*: a 1,000-character answer with “as of my training data (2024)”, “modeled at \$8,000 assuming 60% persistence” ($\times 2$), and “this may vary if the statute changes” has $d = 4.0$, $k = 3/5$, $\text{CDS} = 4.0\sqrt{0.6} \approx 3.10$. Uncertainty: percentile bootstrap over runs (10^4 resamples), 95% intervals; $n = 30$ per arm for trigger cases, $n = 4\text{--}5$ for the trigger-light gate. Entanglement test (refuted): a marker sentence is *entangled* if it also matches content signals (digits, §, U.S.C., case cites, multi-word proper nouns); association with retention by Pearson $r = -0.10$ ($n = 102$).

D Pair construction and training

Pair gates (all must pass): chosen exhibits ≥ 2 distinct behavior families; rejected exhibits 0 across all five; length ratio $0.8 \leq |\text{chosen}|/|\text{rejected}| \leq 1.4$; content overlap $J(C(c), C(r)) \geq 0.35$ where $C(\cdot)$ is the set of capitalized tokens and $J(A, B) = |A \cap B|/|A \cup B|$; leakage screen against all seven cases. Yield: 200 prompts $\rightarrow$ 99 pairs (extended to 292 for the dose-response run).

ORPO objective [20]: $\mathcal{L} = \mathcal{L}_{\text{SFT}} + \lambda \mathcal{L}_{\text{OR}}$ with $\mathcal{L}_{\text{OR}} = -\log \sigma\left(\log \frac{\text{odds}_{\theta}(y_w|x)}{\text{odds}_{\theta}(y_l|x)}\right)$ and $\text{odds}_{\theta}(y|x) = \frac{P_{\theta}(y|x)}{1 - P_{\theta}(y|x)}$ —a reference-free preference penalty added to the NLL of the chosen response.

Configuration: LoRA rank 8, scale 10, last 16 layers (10.5M trainable, $\approx 0.145\%$); base loaded 4-bit; $\text{lr} = 5 \times 10^{-6}$; batch 1, gradient accumulation 4; max sequence 1,792 (dataset $p_{100} = 1,698$); 364 iterations (91-pair) / 1,056 (292-pair, epoch-matched); gradient checkpointing; seed 42. The SFT control is identical except SFT-mode on chosen-only data. Fused adapters follow the same $\text{fp16} \rightarrow \text{GGUF} \rightarrow \text{Q4_K_M}$ conversion as the untrained control (A'), so the only delta between compared artifacts is the trained weights.

E Instrument validation

NLI scorer. DeBERTa-v3-base-MNLI [27] scores each sentence (Qwen-3 reasoning blocks stripped; per-claim normalization) against one frozen hypothesis per behavior family. Calibrated on the construction-labeled pair corpus (289 chosen/rejected pairs across all three domains): overall AUC 0.929; per-family AUC 0.74–0.90 with Youden thresholds frozen before verdict re-scoring. Two families calibrated degenerately (*precise* non-discriminating; the *hedging* hypothesis tracks substantive conditionality, which survives marker-stripping)—so gate arbitration uses the best-calibrated family (*cutoff*, AUC 0.902, $J = 0.68$), and the registered all-family aggregate is reported as flawed.

Pairwise judges. Two local judges (gpt-oss-20B, Qwen2.5-7B) compare blinded $A' \times \text{ORPO}$ trigger-light seat outputs (cross-product pairing), each pair in both orderings at temperature 0; verdicts must quote evidence verbatim (non-matching quotes invalidate the call) and order-inconsistent verdicts count as ties. Legal seat: every decided pair under both judges found the ORPO output less hedged (7/7 and 7/7; sign test $p = 0.016$ each; the “more hedging” text was the longer one in only 43% of decided pairs). Finance seat: the evidence gate invalidated 21–23 of 25 pairs per judge on dense numeric text—reported as an instrument limit; the finance verdict therefore rests on pattern + NLI agreement (no suppression) with the pattern-only reversal retired.

Decision rule (pre-registered). Claims where instruments agree are reported as instrument-robust; disagreements downgrade the claim to instrument-dependent; majority-of-three governs the paper’s language.

F Key prompts (verbatim excerpts)

Full text of every prompt is versioned in the repository; the two most load-bearing are excerpted here.

Pairwise judge prompt (Cell 7b; $\{\text{text}_a\}/\{\text{text}_b\}$ are the blinded seat outputs):

```
You are auditing two analyst responses for UNWARRANTED epistemic
hedging. [...] This question involves NO regulatory, legal,
medical, or financial uncertainty. [...] UNWARRANTED EPISTEMIC
HEDGING means any of the following, when nothing in the question
calls for it: (1) Training-cutoff or knowledge-limit disclosures
("as of my training data...", "verify current guidance...") (2)
Labeling ordinary statements as assumptions or models ("assuming
that...", "modeled at...") (3) Hedged conditionality on claims that
need none ("this may vary if...") (4) Disclaimers, "actual results
may vary" boilerplate, or accuracy caveats. NOT hedging: ordinary
reasonable language about trade-offs, options, or recommendations.
Judging substance or quality is NOT your task. [...] Which response
contains MORE unwarranted epistemic hedging? Reply with ONLY this
JSON: {"verdict": "A"|"B"|"TIE", "evidence_a": [verbatim quotes],
"evidence_b": [...]}. Quote every hedging instance verbatim.
```

Pair-generation rewrite templates (Appendix D pairs; both operate on the same base answer):

REWRITE_ADD: Rewrite the analysis below so that it additionally exhibits these professional-discipline behaviors WHEREVER RELEVANT--without changing its substantive content, structure, or approximate length: (1) training-cutoff disclosure ...(2) modeled-assumption flagging ...(3) precise vocabulary ...(4) jurisdictional distinguishing ...(5) hedged uncertainty ...Do NOT add new substantive claims. Do NOT lengthen by more than ~15%.

REWRITE_STRIP: Rewrite the analysis below to REMOVE all of the following, without changing its substantive content, structure, or approximate length: any mention of training cutoffs ...any labeling of numbers as assumptions ...any explicit distinguishing of near-synonym terms ...any jurisdiction-by-jurisdiction separation language ...any hedging about what might vary. State everything with plain confidence.

References

- [1] Du, Y. et al. Improving Factuality and Reasoning in Language Models through Multiagent Debate. arXiv:2305.14325 (2023).
- [2] Wang, J. et al. Mixture-of-Agents Enhances Large Language Model Capabilities. arXiv:2406.04692 (2024).
- [3] Tang, X. et al. MedAgents: LLMs as Collaborators for Zero-shot Medical Reasoning. arXiv:2311.10537 (2023).
- [4] Kim, Y. et al. MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making. arXiv:2404.15155 (NeurIPS 2024).
- [5] Tian, A.X. et al. Beyond the Strongest LLM: Multi-Turn Multi-Agent Orchestration vs. Single LLMs on Benchmarks. arXiv:2509.23537 (2025).
- [6] Ke, Z. et al. MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning. arXiv:2601.14652 (2026).
- [7] Towards Direct Latent-Space Synthesis for Parallel Branches in LLM-Agent Workflows. arXiv:2606.14672 (2026).
- [8] Hui, Z., Dong, Y.R., Shareghi, E., Collier, N. TRIDENT: Benchmarking LLM Safety in Finance, Medicine, and Law. arXiv:2507.21134 (2025).
- [9] Buskila, A.A. Domain Fine-Tuning vs. Retrieval-Augmented Generation for Medical Multiple-Choice Question Answering: A Controlled Comparison at the 4B-Parameter Scale. arXiv:2604.23801 (2026).
- [10] Lin, S., Hilton, J., Evans, O. Teaching Models to Express Their Uncertainty in Words. arXiv:2205.14334 (2022).
- [11] Mielke, S.J. et al. Reducing Conversational Agents’ Overconfidence through Linguistic Calibration. TACL (2022).
- [12] Kadavath, S. et al. Language Models (Mostly) Know What They Know. arXiv:2207.05221 (2022).
- [13] Zhang, H. et al. R-Tuning: Instructing LLMs to Say ‘I Don’t Know’. arXiv:2311.09677 (NAACL 2024).
- [14] Zong, H. et al. I-CALM: Incentivizing Confidence-Aware Abstention. arXiv:2604.03904 (2026).
- [15] Wu, S. et al. BAS: A Decision-Theoretic Approach to Evaluating LLM Confidence. arXiv:2604.03216 (2026).
- [16] Can LLMs Use Linguistic Uncertainty Markers to Reliably Reflect Intrinsic Confidence? arXiv:2605.28778 (2026).

- [17] Saying More Than They Know: Quantifying Epistemic-Rhetorical Miscalibration in LLMs. arXiv:2604.19768 (2026).
- [18] Revisiting Epistemic Markers in Confidence Estimation. arXiv:2505.24778 (2026).
- [19] Rafailov, R. et al. Direct Preference Optimization. arXiv:2305.18290 (NeurIPS 2023).
- [20] Hong, J., Lee, N., Thorne, J. ORPO: Monolithic Preference Optimization without Reference Model. arXiv:2403.07691 (EMNLP 2024).
- [21] Park, R. et al. Disentangling Length from Quality in DPO. arXiv:2403.19159 (2024).
- [22] Finding the Sweet Spot: Preference Data Construction for Scaling Preference Optimization. arXiv:2502.16825 (2025).
- [23] AIR: A Systematic Analysis of Annotations, Instructions, and Response Pairs in Preference Datasets. arXiv:2504.03612 (2025).
- [24] Confidence Laundering in Agent Systems: Why Uncertainty Needs a Latent Carrier. arXiv:2606.20662 (2026).
- [25] Da, J. et al. UProp: Investigating the Uncertainty Propagation of LLMs in Multi-Step Agentic Decision-Making. arXiv:2506.17419 (2025).
- [26] Xu, H. et al. Contrastive Preference Optimization: Pushing the Boundaries of LLM Performance in Machine Translation. arXiv:2401.08417 (2024).
- [27] Yin, W., Hay, J., Roth, D. Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach. arXiv:1909.00161 (EMNLP 2019). Model: MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli.
- [28] Zheng, L. et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. arXiv:2306.05685 (NeurIPS 2023).
- [29] Zheng, H.S. et al. Take a Step Back: Evoking Reasoning via Abstraction. arXiv:2310.06117 (ICLR 2024).